

187. Physical Medium Dependent (PMD) sublayer and medium, type 800GBASE-ER1-20 and 800GBASE-ER1

187.1 Overview

This clause specifies the 800GBASE-ER1-20 and 800GBASE-ER1 PMDs together with the associated single-mode fiber medium. The optical signals generated by these Physical Medium Dependent (PMD) types are modulated using a dual polarization 16-state quadrature amplitude modulation (DP-16QAM) format suitable for reception by a coherent optical receiver.

When forming a complete Physical Layer, a PMD shall be connected to the 800GBASE-ER1 PMA as shown in Table 187–1.

The 800GBASE-ER1 Physical Layer may optionally include an 800GMII Extender (see Clause 171).

Table 187–1—Physical Layer clauses associated with the 800GBASE-ER1-20 and 800GBASE-ER1 PMD

Associated clause	800GBASE-ER1-20 and 800GBASE-ER1
90—Time Synchronization	Optional
120F—800GAUI-8 C2C	Optional ^a
120G—800GAUI-8 C2M	Optional ^a
170—800 Gb/s RS	Required
170—800GMII ^b	Optional
171—800GMII Extender	Optional ^c
172—800GBASE-R PCS	Required
173—800GBASE-R BM-PMA	Conditional ^d
176—800GBASE-R SM-PMA	Conditional ^d
176C—800GAUI-4 C2C	Optional ^a
176D—800GAUI-4 C2M	Optional ^a
186—800GBASE-ER1 FEC	Required
186—800GBASE-ER1 PMA	Required

^a One or two 800GAUI-n may be instantiated within a 800GBASE-ER1 or 800GBASE-ER1-20 PHY as described in 176B.7.1, except that if an 800GMII Extender is included in the Physical Layer implementation then no 800GAUI-n is permitted in the PHY.

^b The 800GMII is an optional interface. However, if the 800GMII is not implemented, a conforming implementation behaves functionally as though the RS and 800GMII were present.

^c If one or more 800GAUI-n are implemented in the PHY, then an 800GMII Extender is not permitted in the Physical Layer implementation.

^d If one or two 800GAUI-n are implemented in a PHY, 800GBASE-R BM-PMA or SM-PMA sublayers are required according to 176B.7.1.

Figure 187–1 shows the relationship of the PMD and MDI (shown shaded), along with other Ethernet sublayers, to the ISO/IEC Open System Interconnection (OSI) reference model. 800 Gigabit Ethernet is introduced in Clause 169 and the purpose of each PHY sublayer is summarized in 169.2.

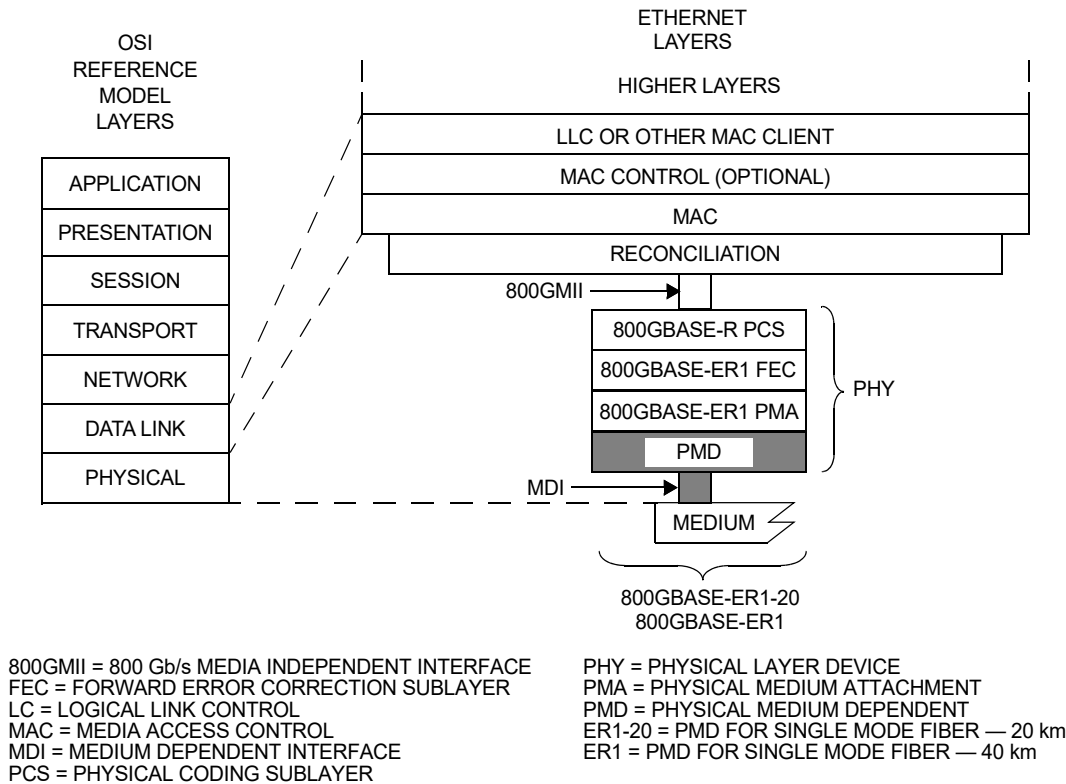


Figure 187–1—800GBASE-ER1-20 and 800GBASE-ER1 PMD relationship to the ISO/IEC Open Systems Interconnection (OSI) reference model and IEEE 802.3 Ethernet model

187.2 Error ratio allocation

With a compliant input signal, a PHY receiver is expected to meet the frame loss ratio specifications in 174A.6.

With a compliant input signal, a PMD receiver is expected to meet the CRC error ratio of 5.7×10^{-11} (see 174A.7), measured at the FEC decoder output.

187.3 PMD service interface

This subclause specifies the services provided by the PMD. The service interface for this PMD is described in an abstract manner and does not imply any particular implementation. The PMD service interface supports the exchange of encoded data between the PMA entity that resides just above the PMD, and the PMD sublayer. The PMD translates the encoded data to and from signals suitable for the specified medium.

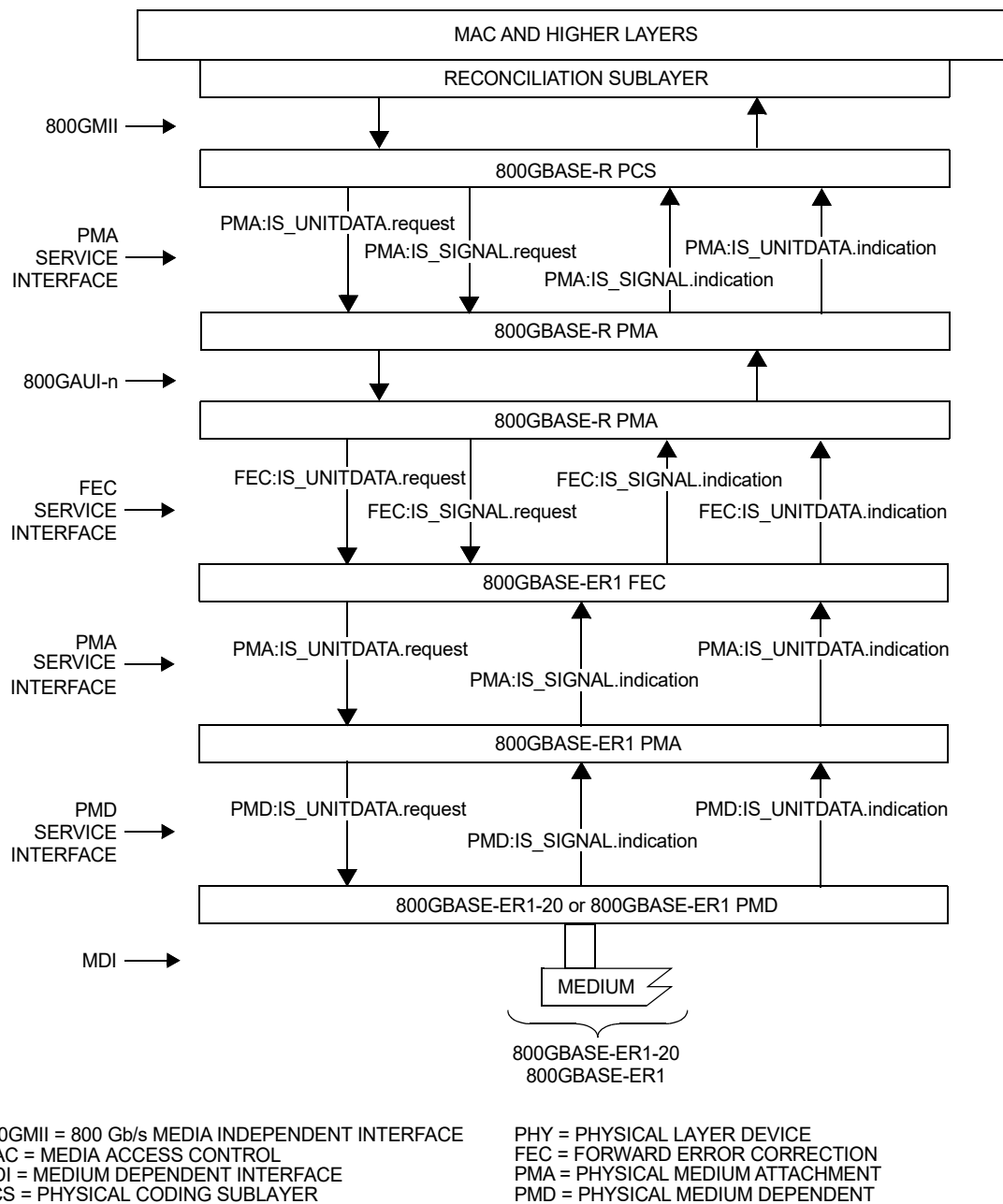


Figure 187-2—800GBASE-ER1 inter-sublayer service interfaces

187.3.1 PMD service interface

The PMD service interface uses the following primitives to exchange signals and status information:

PMD:IS_UNITDATA.request
PMD:IS_UNITDATA.indication
PMD:IS_SIGNAL.indication

187.3.1.1 PMD:IS_UNITDATA.request

The PMD:IS_UNITDATA.request primitive defines the transfer of four analog signals from the 800GBASE-ER1 PMA to the PMD (see 186.3.3.6) that are inputs to the DP-16QAM modulator to generate a single optical signal with orthogonal polarizations (see 187.5.2).

187.3.1.1.1 Semantics of the primitive

PMD:IS_UNITDATA.request(tx_signal_xi, tx_signal_xq, tx_signal_yi, tx_signal_yq)

The PMD:IS_UNITDATA.request primitive conveys four analog signals, representing the in-phase (I) and quadrature (Q) components for each of the polarizations (X and Y), via the tx_signal_xi, tx_signal_xq, tx_signal_yi, and tx_signal_yq parameters.

187.3.1.1.2 When generated

The 800GBASE-ER1 PMA generates PMD:IS_UNITDATA.request continuously.

187.3.1.1.3 Effect of receipt

The effect of receipt of the PMD:IS_UNITDATA.request by the PMD is defined in 187.5.2.

187.3.1.2 PMD:IS_UNITDATA.indication

The PMD:IS_UNITDATA.indication primitive defines the transfer of four analog signals from the PMD to the 800GBASE-ER1 PMA that are the outputs of the coherent DP-16QAM receiver (see 187.5.3).

187.3.1.2.1 Semantics of the primitive

PMD:IS_UNITDATA.indication(rx_signal_xi, rx_signal_xq, rx_signal_yi, rx_signal_yq)

The PMD:IS_UNITDATA.indication primitive conveys four analog signals, representing the in-phase (I) and quadrature (Q) components for each of the polarizations (X and Y), via the rx_signal_xi, rx_signal_xq, rx_signal_yi, rx_signal_yq parameters, respectively.

187.3.1.2.2 When generated

The PMD generates PMD:IS_UNITDATA.indication continuously.

187.3.1.2.3 Effect of receipt

The 800GBASE-ER1 PMA processes the four analog signals as defined in 186.3.4.

187.3.1.3 PMD:IS_SIGNAL.indication(SIGNAL_OK)

The PMD:IS_SIGNAL.indication primitive is generated by the PMD to indicate the status of the receive function via the SIGNAL_OK parameter.

187.3.1.3.1 Semantics of the service primitive

PMD:IS_SIGNAL.indication(SIGNAL_OK)

The SIGNAL_OK parameter corresponds to the variable Global_PMD_signal_detect as defined in 187.5.4. When Global_PMD_signal_detect is 1, SIGNAL_OK shall be assigned the value OK. When

Global_PMD_signal_detect is 0, SIGNAL_OK shall be assigned the value FAIL. When SIGNAL_OK is FAIL, the PMD:IS_UNITDATA_i.indication parameter is undefined.

NOTE—SIGNAL_OK = OK does not guarantee that the rx_signal parameters are known to be good. It is possible for a poor quality link to provide sufficient light for a SIGNAL_OK = OK indication and still not meet the frame loss ratio defined in 187.2.

187.3.1.3.2 When generated

The PMD generates the PMD:IS_SIGNAL.indication primitive whenever there is a change in the value of the SIGNAL_OK parameter.

187.3.1.3.3 Effect of receipt

The effect of receipt by the PMA is defined in 186.3.4.1.

187.4 Delay constraints

The sum of the transmit and receive delays at one end of the link contributed by the PMD including 2 m of fiber in one direction shall be no more than 16 384 bit times (32 pause_quanta or 20.48 ns). A description of overall system delay constraints and the definitions for bit times and pause_quanta can be found in 169.4 and its references.

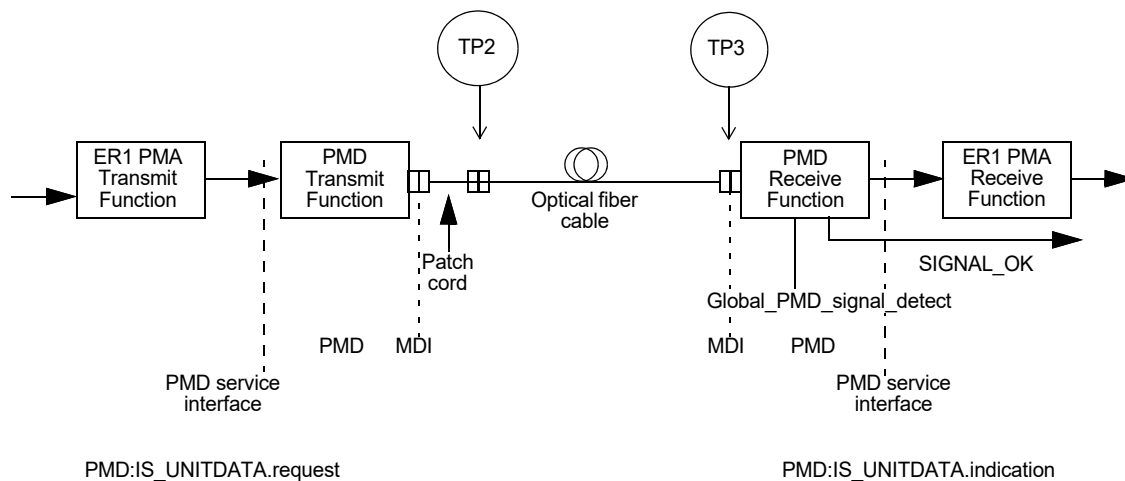
187.5 PMD functional specifications

The PMD performs the Transmit and Receive functions, which convey data between the PMD service interface and the MDI.

187.5.1 Block diagram

A block diagram for the PMD transmit/receive paths is shown in Figure 187–3 and a block diagram of the PMD is shown in Figure 187–4. For purposes of system conformance, the PMD sublayer is standardized at the points described in this subclause. The optical transmit signal is defined at the output end of a single-mode fiber patch cord (TP2), between 2 m and 5 m in length. Unless specified otherwise, all transmitter measurements and tests defined in 187.8 and 187.9 are made at TP2. The optical receive signal is defined at the output of the fiber optic cabling (TP3) at the MDI (see 187.7). Unless specified otherwise, all receiver measurements and tests defined in 187.8 are made at TP3.

While TP2 and the patch cord for transmitter measurements and tests are shown in Figure 187–3, they are not part of the medium. The transmit MDI is connected to the medium for use, or to the patch cord and TP2 for transmitter measurements and tests.



For clarity, only one direction of transmission is shown

Figure 187-3—Block diagram for 800GBASE-ER1-20 and 800GBASE-ER1 transmit/receive

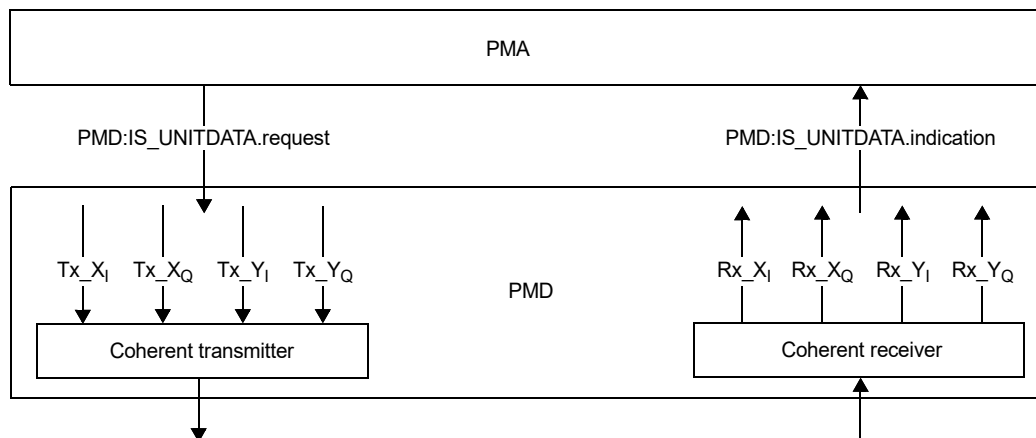


Figure 187-4—Block diagram of 800GBASE-ER1-20 and 800GBASE-ER1 PMD

187.5.2 PMD transmit function

The PMD transmit function shall convert the four analog streams (Tx_X_I , Tx_X_Q , Tx_Y_I , Tx_Y_Q) from the PMA passed across the PMD service interface via the `PMD:IS_UNITDATA.request` primitive (see 187.3.1) into a single optical signal with orthogonal polarizations modulated using 16QAM and deliver them to the MDI, all according to the transmit optical specifications in this clause.

The analog signals use any of the mappings in Table 187–2.

Table 187–2—Allowed analog signal to modulator input mappings

Mapping	Tx_X _I	Tx_X _Q	Tx_Y _I	Tx_Y _Q
1	X _I	X _Q	Y _I	Y _Q
2	X _Q	X _I	Y _Q	Y _I
3	X _I	X _Q	Y _Q	Y _I
4	X _Q	X _I	Y _I	Y _Q
5	Y _I	Y _Q	X _I	X _Q
6	Y _Q	Y _I	X _Q	X _I
7	Y _Q	Y _I	X _I	X _Q
8	Y _I	Y _Q	X _Q	X _I

187.5.3 PMD receive function

The PMD receive function shall convert the composite optical signal received from the MDI into four analog streams (Rx_X_I, Rx_X_Q, Rx_Y_I, Rx_Y_Q) passed across the PMD service interface to the PMA via the PMD:IS_UNITDATA.indication primitive (see 187.3.1.2), all according to the receive optical specifications in this clause.

The four analog streams carry a combination of the transmitting PMA Tx_X_I, Tx_X_Q, Tx_Y_I, and Tx_Y_Q signals used by the transmitting PMD to generate the DP-16QAM symbols.

187.5.4 Global PMD signal detect function

The variable Global_PMD_signal_detect is a global indicator of the presence of an optical signal. For the PMD, the Global_PMD_signal_detect is set to PMD_signal_detect (see 187.5.5).

The state of Global_PMD_signal_detect is conveyed to PMD client sublayers via the PMD service interface (see 187.3).

187.5.5 PMD signal detect function

The PMD signal detect function is used by the PMD to indicate sufficient optical power is detected at the receiver input. A receiver shall set the variable PMD_signal_detect to either 0 or 1 according to the criteria in Table 187–6.

Various implementations of the signal detect function are permitted by this standard. This standard imposes no response time requirements on the generation of PMD_signal_detect.

Implementations need to provide adequate margin between the input optical power level at which PMD_signal_detect is set to 1 and the inherent noise level of the PMD, including the effects of crosstalk, power supply noise, etc.

Table 187–3—Signal detection criteria

Receive conditions	PMD_signal_detect
Average optical power at TP3 ≤ -19.5 dBm	0
[The input signal is a compliant 800GBASE-ER1-20 or 800GBASE-ER1 signal] ^a AND [Optical power at TP3 $\geq$ average receive power (min) in Table 187–6]	1
All other conditions	Unspecified, may be 0 or 1

^a This is a condition for testing of the receiver, not a requirement for the receiver to detect that the signal is valid.

187.5.6 PMD reset function

If the variable PMD_reset is asserted, the PMD shall be reset as defined in 45.2.1.1.1.

187.5.7 Global PMD transmit disable function (optional)

The global PMD transmit disable function is optional and allows the optical transmitter to be disabled.

When the Global_PMD_transmit_disable variable is set to one, this function shall turn off the optical transmitter so that it meets the requirements of the average launch power of the OFF transmitter in Table 187–5.

If a PMD fault is detected (see 187.5.8), then the PMD may set the Global_PMD_transmit_disable variable to one, turning off the optical transmitter.

187.5.8 PMD fault function (optional)

If the PMD has detected a local fault on any of the transmit or receive paths, the PMD shall set PMD_fault variable to one.

187.5.9 PMD transmit fault function (optional)

If the PMD has detected a local fault on the transmit path, the PMD shall set the PMD_transmit_fault variable to one.

187.5.10 PMD receive fault function (optional)

If the PMD has detected a local fault on the receive path, the PMD shall set the PMD_receive_fault variable to one.

187.5.11 Autonomous path startup (APSU) functions

The 800GBASE-ER1 PMD supports APSU via the 800GBASE-ER1 FEC sublayer (see 186.2.6).

187.6 PMD to MDI optical specifications

The operating range for the PMD is defined in Table 187–4. A compliant PMD operates over single mode fibers meeting the specifications in Table 187–9. A PMD that exceeds the operating range requirement while meeting all other optical specifications is considered compliant (e.g., a 800GBASE-ER1 PMD that could operate over 45 km would meet the operating range requirement of 2 m to 40 km).

Table 187–4—Operating range

PMD type	Required operating range
800GBASE-ER1-20	2 m to 20 km
800GBASE-ER1	2 m to 40 km

187.6.1 Transmitter optical specifications

The transmitter shall meet the specifications defined in Table 187–5 per the definitions in 187.8.

Table 187–5—800GBASE-ER1-20 and 800GBASE-ER1 transmit characteristics

Description	Reference	800GBASE-ER1-20	800GBASE-ER1	Unit
Signaling rate (range)	—	118.203351 ± 20 ppm		GBd
Modulation format	—	DP-16QAM		—
Average launch power (max)	187.8.2	–5	–1	dBm
Average launch power (min) for ETCC ≤ 1 dB for 1 < ETCC ≤ 2.5 dB	187.8.2	–11 –12 + ETCC	–5 –6 + ETCC	dBm
Carrier frequency (range)	187.8.3	193.7 ± 1.8 GHz		THz
Power difference between X and Y polarizations (max)	187.8.4	1		dB
Skew between X and Y polarizations (max)	187.8.5	5		ps
ETCC (max)	187.8.6	2.5		dB
Instantaneous I-Q offset per polarization (max)	187.8.7	–20		dB
Mean I-Q offset per polarization (max)	187.8.8	–26		dB
I-Q amplitude imbalance (mean)	187.8.9	1		dB
I-Q phase error magnitude (max)	187.8.10	5		deg
I-Q quadrature skew (max)	187.8.11	0.75		ps
Transmitter OSNR in a 12.5 GHz bandwidth (min)	187.8.12	35		dB

Table 187–5—800GBASE-ER1-20 and 800GBASE-ER1 transmit characteristics (*continued*)

Description	Reference	800GBASE-ER1-20	800GBASE-ER1	Unit
Average launch power of OFF transmitter (max)	—	–20		dBm
Transmitter reflectance ^a (max)	—	–20		dB
RIN average (max)	—	–145		dB/Hz
RIN peak (max)	—	–140		dB/Hz
Laser linewidth (max)	—	1		MHz
Tx clock phase noise: phase noise mask frequency (max)	187.8.13			
1×10^4		–100		dBc/Hz
4×10^5		–132		dBc/Hz
1×10^6		–136		dBc/Hz
$\leq 1 \times 10^7$		–146		dBc/Hz
Tx clock phase noise: total integrated random jitter, RMS (max)	187.8.14	0.015		UI
Tx clock phase noise: total periodic jitter, peak-to-peak (max)	187.8.15	0.03		UI

^a Transmitter reflectance is defined looking into the transmitter.

187.6.2 Receive optical specifications

The receiver shall meet the specifications defined in Table 187–6 per the definitions in 187.8.

Table 187–6—800GBASE-ER1-20 and 800GBASE-ER1 receive characteristics

Description	Reference	800GBASE-ER1-20	800GBASE-ER1	Unit
Signaling rate (range)	—	118.203351 ± 20 ppm		GBd

Table 187–6—800GBASE-ER1-20 and 800GBASE-ER1 receive characteristics (*continued*)

Description	Reference	800GBASE-ER1-20	800GBASE-ER1	Unit
Modulation format	—	DP-16QAM		—
Carrier frequency (range)	—	193.7 ± 1.8 GHz		THz
Damage threshold ^a	—	2		dBm
Average receive power tolerance (max)	187.8.16	−1		dBm
Average receive power tolerance (min) for $ETCC \leq 1$ dB for $1 < ETCC \leq 2.5$ dB	187.8.16	−18 −19 + ETCC		dBm
Receiver sensitivity (max) ^b for $ETCC \leq 1$ dB for $1 < ETCC \leq 2.5$ dB	187.8.17	−18.5 −19.5 + ETCC	−19 −20 + ETCC	dBm
Receiver reflectance (max)	—	20		dB
Frequency offset between received carrier and local oscillator (max)	—	± 1.8		GHz
Polarization dependent loss (max)	—	1.5		dB
Polarization rotation speed (max)	—	50		krad/s

^a The receiver shall be able to tolerate, without damage, continuous exposure to an optical input signal having this average power level. The receiver does not have to operate correctly at this input power.

^b Receiver sensitivity is optional.

187.6.3 Illustrative link power budget

An illustrative power budget and penalties for 800GBASE-ER1-20 and 800GBASE-ER1 channels are shown in Table 187–7.

Table 187–7—800GBASE-ER1-20 and 800GBASE-ER1 illustrative link power budget

Parameter	800GBASE-ER1-20	800GBASE-ER1	Unit
Power budget	7.5	14	dB
Operating distance	20	40	km
Channel insertion loss ^a	7	12	dB
Maximum discrete reflectance	−27		dB
Allocation for penalties ^b	0.5	1	dB
Additional insertion loss allowed	0	1	dB

^a The channel insertion loss is calculated using the maximum distance specified in Table 187–4 and cabled optical fiber attenuation of 0.25 dB/km at 1550 nm plus an allocation for connection and splice loss given in 187.7.2.1.

^b Link penalties are used for link budget calculations. They are not requirements and are not meant to be tested.

187.7 Optical channel characteristics

The fiber optic cabling model is shown in Figure 187–5.

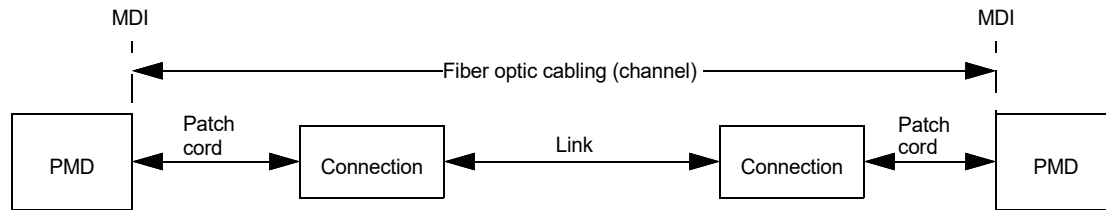


Figure 187–5—Fiber optic cabling model

The optical channel characteristics shall meet the specifications defined in Table 187–8. The fiber optic cabling consists of one or more sections of fiber optic cable and any intermediate connections required to connect sections together. A channel may contain additional connectors as long as the optical characteristics of the channel, such as attenuation, dispersion, reflections, and polarization mode dispersion meet the specifications. Insertion loss measurements of installed fiber cables are made in accordance with IEC 61280-4-2 one-cord reference method. The fiber optic cabling model (channel) defined here applies to each simplex fiber that makes up the duplex fiber link segment. The term channel is used here for consistency with generic cabling standards.

Table 187–8—Optical channel characteristics

Description	800GBASE-ER1-20	800GBASE-ER1	Unit
Operating distance (max)	20	40	km
Channel insertion loss ^{a,b} (max)	7	12	dB
Channel insertion loss (min)	0		dB
Chromatic dispersion (max)	400	800	ps/nm
Chromatic dispersion (min)	0		ps/nm
Differential group delay (max) ^c	7	10	ps
Optical return loss (min)	24		dB

^a These channel insertion loss values include cable, connectors, and splices.

^b At 1550 nm.

^c Differential Group Delay (DGD) is the time difference at reception between the fractions of a pulse that were transmitted in the two principal states of polarization of an optical signal. DGD_max is the maximum differential group delay that the system is required to tolerate.

187.7.1 Optical fiber cable

The optical fiber cable requirements are satisfied by cables containing ITU-T G.652.B (dispersion unshifted), type G.652.D (low water peak, dispersion unshifted), or type G.657.A1, or type G.657.A2 (bend insensitive) fibers, or the requirements in Table 187–9 where they differ.

Table 187–9—Optical fiber and cable characteristics

Description	Value	Unit
Nominal fiber specification wavelength	1550	nm
Cabled optical fiber attenuation (max)	0.25 ^a	dB/km
Zero dispersion wavelength (λ_0)	$1300 \leq \lambda_0 \leq 1324$	nm
Dispersion slope (max) (S_0)	0.092	ps/(nm ² × km)

^a The 0.25 dB/km attenuation provided for outside plant cable has been conservatively derived using the link attenuation model described in ITU-T G.652 Appendix I using attenuation coefficient values from ITU-T G.695 Appendix I or equivalent.

187.7.2 Optical fiber connection

An optical fiber connection, as shown in Figure 187–5, consists of a mated pair of optical connectors.

187.7.2.1 Connection insertion loss

The maximum link distance is based on an allocation of 2 dB total connection and splice loss. For example, this allocation supports four connections with an average insertion loss per connection of 0.5 dB. Connections with different loss characteristics may be used provided the requirements of Table 187–8 are met.

187.7.3 Medium Dependent Interface (MDI)

The PMD is coupled to the fiber optic cabling via one fiber per direction at the MDI, being the interface between the PMD and the medium. Examples of an MDI include the following:

- Connectorized fiber pigtail
- PMD receptacle

When the MDI is a connector plug and receptacle connection, it shall meet the interface performance specifications of IEC 61753-1 and IEC 61753-021-02.

NOTE—Transmitter compliance testing is performed at TP2 as defined in 187.5.1, not at the MDI.

187.8 Definition of optical parameters and measurement methods

All transmitter optical measurements shall be made through a short patch cable, between 2 m and 5 m in length, unless otherwise specified.

187.8.1 Test patterns for optical parameters

While compliance is to be achieved in normal operation, specific test patterns are defined for measurement consistency and to enable measurement of some parameters. Table 187–11 gives the test patterns to be used in each measurement, unless otherwise specified, and also lists references to the subclauses in which each

parameter is defined. Any of the test patterns given for a particular test in Table 187–11 may be used to perform that test. The test patterns used in this clause are shown in Table 187–10.

Table 187–10—Test patterns

Pattern	Pattern description	Defined in
5	Scrambled idle test pattern encoded by the 800GBASE-ER1 FEC sublayer FEC	172.2.4, 186.2.3.12
7	Valid 800GBASE-R signal encoded by the 800GBASE-ER1 FEC sublayer	172.2.4, 186.2.3.12
8	PRBS31 encoded by the 800GBASE-ER1 FEC sublayer	186.2.3.12

Table 187–11—Mapping of parameters to test-patterns and related subclauses

Parameter	Pattern	Related subclause
Average channel output power	5, 7, or 8	187.8.2
Carrier frequency (range)	5, 7, or 8	187.8.3
Power difference between X and Y polarizations	8	187.8.4
Skew between X and Y polarizations	8	187.8.5
ETCC	8	187.8.6
Instantaneous I-Q offset per polarization	8	187.8.7
Mean I-Q offset per polarization	8	187.8.8
I-Q amplitude imbalance	8	187.8.9
I-Q phase error magnitude	8	187.8.10
I-Q quadrature skew	8	187.8.11
Transmitter in-band OSNR	8	187.8.12
Tx clock phase noise: phase noise mask frequency	8	187.8.13
Tx clock phase noise: total integrated random jitter	8	187.8.14
Tx clock phase noise: total periodic jitter	8	187.8.15
Average receive power tolerance	5, 7, or 8	187.8.16
Receiver sensitivity	5, 7, or 8	187.8.17

187.8.2 Average channel output power

The average channel output power shall be within the limits given in Table 187–5 if measured using the methods given in IEC 61280-1-1. The average optical power is measured using the test pattern defined in Table 187–11, per the test setup in Figure 53–6.

187.8.3 Carrier frequency (range)

The carrier frequency (range) shall be within the range given in Table 187–5 if measured per IEC 61280-1-3. The transmitter is modulated using the test pattern defined in Table 187–11.

187.8.4 Power difference between X and Y polarizations

The power difference between X and Y polarizations, as defined in Recommendation ITU-T G.698.2, shall be within the limit given in Table 187–5.

187.8.5 Skew between X and Y polarizations

The skew between the X and Y polarizations, as defined in Recommendation ITU-T G.698.2, shall be within the limit given in Table 187–5.

187.8.6 Extended transmitter constellation closure (ETCC)

ETCC is the metric used to define the quality of a 800 Gb/s DP-16QAM transmitter. The ETCC is computed using the test setup and calculation defined in Annex 185A and the parameter values listed in Table 187–12, Table 187–13, and Table 187–14.

The ETCC shall be within the limit given in Table 187–5 if measured using the methods specified.

187.8.6.1 Coherent detector front-end

Parameters for the coherent detector front-end shall meet the values listed in Table 187–12.

Table 187–12—Coherent detector front-end parameters

Description	Value	Unit
3 dB bandwidth (min)	65	GHz
Effective number of bits (<i>ENOB</i>) (min)	5	bits
Oversampling ratio (min)	1.25	—
Local oscillator linewidth (max)	100	kHz
BER_{ref}	2×10^{-2}	—

187.8.6.2 Calibrated front-end sampling

The post calibration residuals for the coherent detector front-end shall meet the values listed in Table 187–13.

Table 187–13—Coherent front-end post calibration residuals

Description	Value	Unit
X-Y gain error (max)	0.2	dB
Skew between X-Y polarizations (max)	0.5	ps
I-Q phase error (max)	2	degree
I-Q gain error (max)	0.2	dB
I-Q skew (max)	0.5	ps
Carrier frequency offset (max)	0.1	GHz
Frequency response magnitude variation (max)	1	dB
Frequency response phase variation (max)	0.3	rad
White noise power spectral density variation (max)	0.1	dB
Signal to receiver noise ratio after noise whitening (min)	20	dB

187.8.6.3 Digital signal processing calculation

The digital signal processing parameters for the reference equalizers shall meet the values listed in Table 187–14.

Table 187–14— Digital signal processing parameters

Description	Value
Number of reference equalizer taps	31
Number of reference post-equalizer taps	5

187.8.7 Instantaneous I-Q offset per polarization

The instantaneous I-Q offset per polarization shall be within the limit given in Table 187–5. The instantaneous I-Q offset per polarization is the maximum value per polarization and is calculated as $I\text{-}Q_{\text{offset}} = 10\log_{10}[(I_{\text{mean}}^2 + Q_{\text{mean}}^2)/P_{\text{signal}}]$ with a measurement interval of 1 μs .

187.8.8 Mean I-Q offset per polarization

The mean I-Q offset per polarization shall be within the limit given in Table 187–5. The mean I-Q offset per polarization is the mean value per polarization and is calculated as $I\text{-}Q_{\text{offset}}(\text{mean}) = 10\log_{10}[(I_{\text{mean}}^2 + Q_{\text{mean}}^2)/P_{\text{signal}}]$.

187.8.9 I-Q amplitude imbalance

The I-Q amplitude imbalance is the amplitude difference in dB of the in-phase component I and quadrature component Q of the signal and shall be within the limit given in Table 187–5.

187.8.10 I-Q phase error magnitude

The I-Q phase error magnitude is the phase difference of the in-phase component I and quadrature component Q of the signal, measured relative to local oscillator and shall be within the limit given in Table 187–5.

187.8.11 I-Q quadrature skew

The I-Q quadrature skew is the relative skew between the in-phase component I and quadrature component Q of the signal and shall be within the limit given in Table 187–5.

187.8.12 Transmitter in-band OSNR

The transmitter in-band OSNR shall be within the limit given in Table 187–5. The transmitter in-band OSNR is defined as the ratio of the total signal power in transmitted channel to the highest noise power density (in a 12.5 GHz resolution bandwidth) in the range of the central frequency.

187.8.13 Tx clock phase noise: phase noise mask frequency

The Tx clock phase noise: phase noise mask frequency, as defined in OIF-800ZR-01.0, Implementation Agreement for 800ZR Coherent Interfaces, section 6.1.213a, shall be within the limits given in Table 187–5.

187.8.14 Tx clock phase noise: total integrated random jitter

The Tx clock phase noise: total integrated random jitter, as defined in OIF-800ZR-01.0, Implementation Agreement for 800ZR Coherent Interfaces, section 6.1.213b, shall be within the limits given in Table 187–5.

187.8.15 Tx clock phase noise: total periodic jitter

The Tx clock phase noise: total periodic jitter, as defined in OIF-800ZR-01.0, Implementation Agreement for 800ZR Coherent Interfaces, section 6.1.213c, shall be within the limits given in Table 187–5.

187.8.16 Average receive power tolerance

The average receive power tolerance defines the range of average receiver input power at TP3 over which the CRC ratio requirement in 187.2 is met. Average receiver power tolerance is measured using the patterns listed in Table 187–10.

The conformance test signal meets the requirements for an 800GBASE-ER1 transmitter, with ETCC equal to the maximum value specified in Table 187–5, followed by a channel with the following impairments: polarization dependent loss and polarization rotation speed as specified in Table 187–6, and differential group delay and dispersion as specified in Table 187–8.

The average receive power tolerance shall meet the limits given in Table 187–6.

187.8.17 Receiver sensitivity

Receiver sensitivity is an optional parameter defined as the lowest average receiver input power at TP3 at which the CRC error ratio requirement in 187.2 is met. This does not have to be met in the presence of

impairments from the link, which are addressed separately in the allocation for penalties in Table 187–7. Receiver sensitivity is measured using the patterns listed in Table 187–10.

The conformance test signal meets the requirements for an 800GBASE-ER1 transmitter with ETCC equal to the maximum value specified in Table 187–5, followed by an optical attenuator.

187.9 Safety, installation, environment, and labeling

187.9.1 General safety

All equipment subject to this clause shall conform to the general safety requirements in J.2.

187.9.2 Laser safety

The optical transceivers shall conform to Hazard Level 1 laser requirements as defined in IEC 60825-1 and IEC 60825-2, under any condition of operation. This includes single fault conditions whether coupled into a fiber or out of an open bore.

Conformance to additional laser safety standards may be required for operation within specific geographic regions.

Laser safety standards and regulations require that the manufacturer of a laser product provide information about the product’s laser, safety features, labeling, use, maintenance, and service. This documentation explicitly defines requirements and usage restrictions on the host system necessary to meet these safety certifications.²¹

187.9.3 Installation

It is recommended that proper installation practices, as defined by applicable local codes and regulation, be followed in every instance in which such practices are applicable.

187.9.4 Environment

Normative specifications in this clause shall be met by a system integrating a 800GBASE-ER1-20 or 800GBASE-ER1 PMD over the life of the product while the product operates within the manufacturer’s range of environmental, power, and other specifications.

It is recommended that manufacturers indicate in the literature associated with the PHY the operating environmental conditions to facilitate selection, installation, and maintenance.

It is recommended that manufacturers indicate, in the literature associated with the components of the optical link, the distance and operating environmental conditions over which the specifications of this clause are met.

187.9.5 Electromagnetic emission

A system integrating a 800GBASE-ER1-20 or 800GBASE-ER1 PMD shall comply with applicable local and national codes for the limitation of electromagnetic interference.

²¹A host system that fails to meet the manufacturer’s requirements and/or usage restrictions may emit laser radiation in excess of the safety limits of one or more safety standards. In such a case, the host manufacturer is required to obtain its own laser safety certification.

187.9.6 Temperature, humidity, and handling

The optical link is expected to operate over a reasonable range of environmental conditions related to temperature, humidity, and physical handling (such as shock and vibration). Specific requirements and values for these parameters are considered to be beyond the scope of this standard.

187.9.7 PMD labeling requirements

It is recommended that each PHY (and supporting documentation) be labeled in a manner visible to the user, with at least the applicable safety warnings and the applicable port type designation (e.g., 800GBASE-ER1).

Labeling requirements for Hazard Level 1 lasers are given in the laser safety standards referenced in 187.9.2.

187.10 Management variables

PMD control and status variables intended to be accessible via a management system are listed in Table 187–15 and Table 187–16.

If the MDIO interface (see Clause 45) is implemented, PMD control and status variables shall be mapped to MDIO registers as shown in Table 187–15 and Table 187–16.

If the MDIO interface is not implemented, an alternate mechanism to access management variables shall be provided.

Table 187–15—PMD control variables and MDIO mapping

Control variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
PMD reset	187.5.6	1.0.15	45.2.1.1.1
Global PMD transmit disable	187.5.7	1.9.0	45.2.1.8.7

Table 187–16—PMD status variables and MDIO mapping

Status variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
PMD fault	187.5.8	1.1.7	45.2.1.2.3
PMD transmit fault	187.5.9	1.8.11	45.2.1.7.4
PMD receive fault	187.5.10	1.8.10	45.2.1.7.5
Global PMD signal detect	187.5.4	1.10.0	45.2.1.9.7
PMD signal detect	187.5.5	1.10.1	45.2.1.9.6

187.11 Protocol implementation conformance statement (PICS) proforma for Clause 187, Physical Medium Dependent (PMD) sublayer and medium, type 800GBASE-ER1-20 and 800GBASE-ER1²²

187.11.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 187, Physical Medium Dependent (PMD) sublayer and medium, type 800GBASE-ER1-20 and 800GBASE-ER1, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

187.11.2 Identification

187.11.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1,3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

187.11.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3dj-202x, Clause 187, Physical Medium Dependent (PMD) sublayer and medium, type 800GBASE-ER1-20 and 800GBASE-ER1
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No [] Yes [] (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3dj-202x.)	

Date of Statement	
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²²Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

187.11.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
ER1	800GBASE-ER1	187.6	Device supports requirements for 800GBASE-ER1 PHY	O.1	Yes [] No []
TP1	Reference point TP1 exposed and available for testing	187.5.1	This point may be made available for use by implementers to certify component conformance	O	Yes [] No []
TP4	Reference point TP4 exposed and available for testing	187.5.1	This point may be made available for use by implementers to certify component conformance	O	Yes [] No []
DC	Delay constraints	187.4	Device conforms to delay constraints	M	Yes []
MD	MDIO capability	187.10	Registers and interface supported	O	Yes [] No []

187.11.4 PICS proforma tables for Physical Medium Dependent (PMD) sublayer and medium, type 800GBASE-ER1

187.11.4.1 PMD functional specifications

Item	Feature	Subclause	Value/Comment	Status	Support
F1	Compatible with 800GBASE-R PCS, 800GBASE-ER1 FEC, and 800GBASE-ER1 PMA	187.1		M	Yes []
F2	Integration with management functions	187.1		O	Yes [] No []
F3	Error ratio allocation	187.2	Meets error ratio allocation specified in 187.2	M	Yes []
F4	Transmit function	187.5.2	Conveys analog data from PMD service interface to MDI	M	Yes []
F5	Receive function	187.5.3	Conveys analog data from PMD service interface to MDI	M	Yes []
F6	Global signal detect	187.5.4	Report to the PMD service interface the message PMD:IS_SIGNAL.indication (SIGNAL_OK)	O	Yes [] No []
F7	PMD reset function	187.5.6	Resets the PMD sublayer	O	Yes [] No []

Item	Feature	Subclause	Value/Comment	Status	Support
F8	Global PMD transmit disable function	187.5.7	Disables the optical transmitter with the Global_PMD_transmit_disable variable	O	Yes [] No []
F9	PMD fault function	187.5.8	Sets PMD_fault to one if a local fault is detected	O	Yes [] No []
F10	PMD transmit fault function	187.5.9	Sets PMD_transmit_fault to one if a local fault is detected	O	Yes [] No []
F11	PMD receive fault function	187.5.10	Sets PMD_receive_fault to one if a local fault is detected	O	Yes [] No []

187.11.4.2 PMD to MDI optical specifications for 800GBASE-ER1

Item	Feature	Subclause	Value/Comment	Status	Support
ER1	Transmitter meets specifications in Table 187–5	187.6.1	Per definitions in 187.8	M	Yes []
ER2	Receiver meets specifications in Table 187–6	187.6.2	Per definitions in 187.8	M	Yes []

187.11.4.3 Optical measurement methods

Item	Feature	Subclause	Value/Comment	Status	Support
OM1	Measurement cable	187.8	2 m to 5 m in length	M	Yes []
OM2	Average channel output power	187.8.2	Per IEC 61280-1-1	M	Yes []
OM3	Carrier frequency (range)	187.8.3		M	Yes []
OM4	Power difference between X and Y polarizations	187.8.4		M	Yes []
OM5	Skew between X and Y polarizations	187.8.5		M	Yes []
OM6	ETCC	187.8.6		M	Yes []
OM7	Instantaneous I-Q offset per polarization	187.8.7		M	Yes []
OM8	Mean I-Q offset per polarization	187.8.8		M	Yes []
OM9	I-Q amplitude imbalance	187.8.9		M	Yes []
OM10	I-Q phase error magnitude	187.8.10		M	Yes []
OM11	I-Q quadrature skew	187.8.11		M	Yes []
OM12	Transmitter in-band OSNR	187.8.12		M	Yes []

Item	Feature	Subclause	Value/Comment	Status	Support
OM13	Tx clock phase noise: phase noise mask frequency	187.8.13		M	Yes []
OM14	Tx clock phase noise: total integrated random jitter	187.8.14		M	Yes []
OM15	Tx clock phase noise: total periodic jitter	187.8.15		M	Yes []
OM16	Average receive power tolerance	187.8.16		M	Yes []

187.11.4.4 Environmental specifications

Item	Feature	Subclause	Value/Comment	Status	Support
ES1	General safety	187.9.1	Conforms to J.2	M	Yes []
ES2	Laser safety	187.9.2	Conforms to Hazard Level 1 laser requirements defined in IEC 60825-1 and IEC 60825-2	M	Yes []
ES3	Temperature, humidity, and handling	187.9.6	Complies with applicable local and national codes for the limitation of electromagnetic interference	M	Yes []

187.11.4.5 Characteristics of the fiber optic cabling and MDI

Item	Feature	Subclause	Value/Comment	Status	Support
OC1	Fiber optic cabling	187.7.1	Meets the requirements in Table 187–9	M	Yes []
OC2	MDI requirements	187.7.3	Meets IEC 61753-1 and IEC 61753-021-02	INS:M	Yes [] N/A []

187.11.4.6 Management

Item	Feature	Subclause	Value/Comment	Status	Support
M1	Alternate access to management variables is provided	187.10		!MD:M	Yes [] N/A []
M2	Mapping of MDIO control bits and MDIO status bits	187.10		MD:M	Yes [] N/A []